5G Network Simulation using Ella Core and UERANSIM

Student Details

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1. Introduction

This experiment demonstrates the simulation of a 5G network using Ella Core and UERANSIM. The objective is to establish communication between the 5G core network and user equipment (UE) and analyze the behavior of the system during connection attempts.

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2. Tools Used

- Ella Core (attempted)
- UERANSIM
- Ubuntu 22.04 (Virtual Machine)
- Docker

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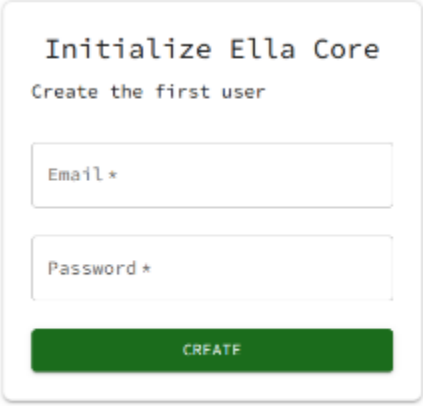
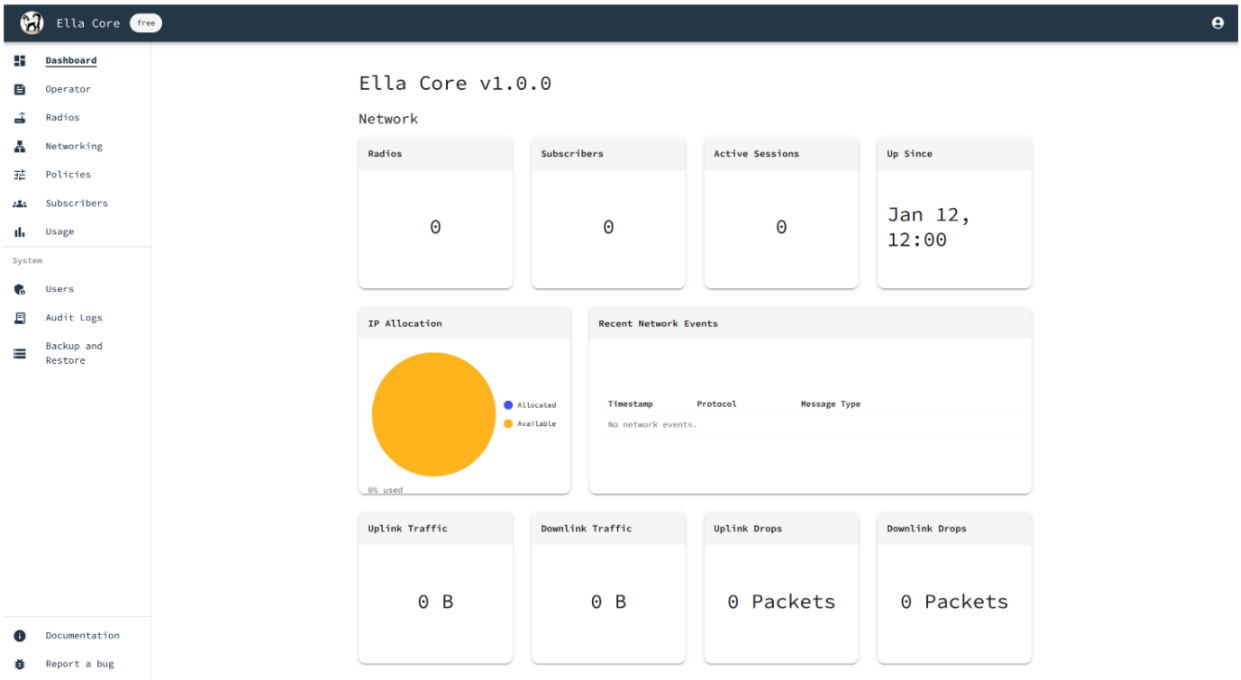
3. Procedure

1. Installed Ella Core and UERANSIM on the system.
2. Initialized Ella Core using Docker containers.
3. Accessed the Ella Core web dashboard.
4. Created a subscriber with IMSI and authentication parameters.
5. Installed and built UERANSIM.
6. Configured and launched gNB (5G base station simulator).
7. Configured and launched UE (User Equipment simulator).
8. Verified successful registration and session establishment.
9. Cleaned up the setup.

4. Screenshots

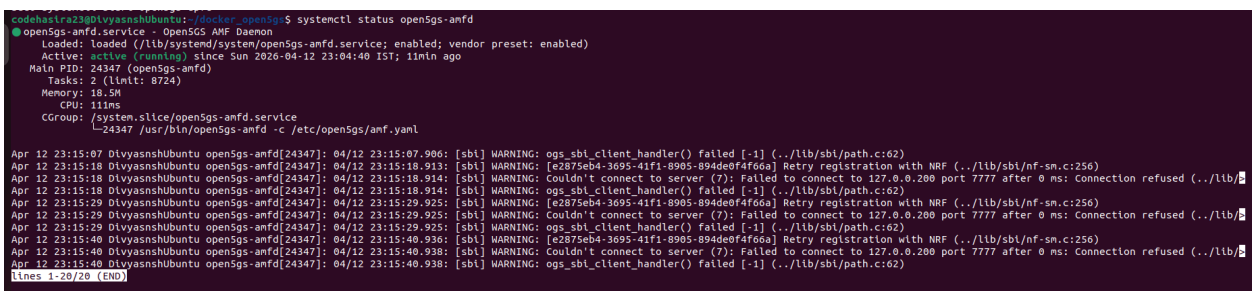
4.1 Ella Core Initialization Page





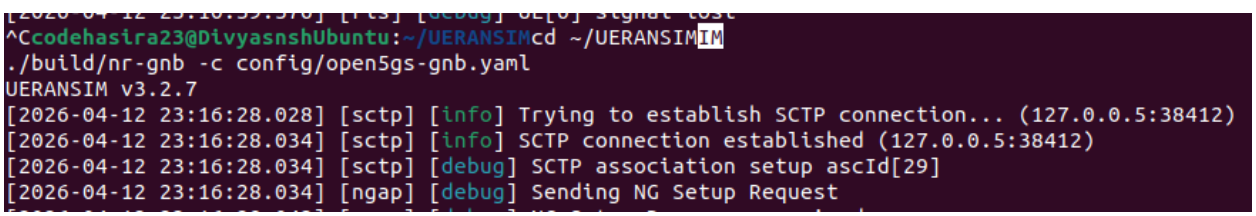
Subscribers

4.2 Open5GS AMF Service Status



Caption: AMF (Access and Mobility Function) service running in the 5G core network.

4.3 Connection with 5G Radio Simulator (gNB)



Caption: gNB attempting SCTP connection with core network.

4.4 Connection with User Equipment Simulator (UE)

```
codehasira23@DivyasnshUbuntu:~/UERANSIM$ cd ~/UERANSIMIM
./build/nr-ue -c config/open5gs-ue.yaml
UERANSIM v3.2.7
[2026-04-12 23:16:40.770] [nas] [info] UE switches to state [MM-DEREGISTERED/PLMN-SEARCH]
[2026-04-12 23:16:40.771] [rrc] [debug] New signal detected for cell[1], total [1] cells in coverage
[2026-04-12 23:16:40.772] [nas] [info] Selected plmn[999/70]
[2026-04-12 23:16:40.772] [rrc] [info] Selected cell plmn[999/70] tac[1] category[SUITABLE]
[2026-04-12 23:16:40.772] [nas] [info] UE switches to state [MM-DEREGISTERED/PS]
[2026-04-12 23:16:40.773] [nas] [info] UE switches to state [MM-DEREGISTERED/NORMAL-SERVICE]
```

Caption: UE detecting signal and attempting cell selection.

4.5 Terminal Window Output

```
[2026-04-12 23:17:45.836] [nas] [info] UE switches to state [CM-IDLE]
[2026-04-12 23:17:55.846] [nas] [debug] NAS timer[3511] expired [1]
[2026-04-12 23:17:55.846] [nas] [debug] Initial registration required due to [T3511-EXPIRY-IN-ATT-REG]
[2026-04-12 23:17:55.846] [nas] [debug] UAC access attempt is allowed for identity[0], category[MO_sig]
[2026-04-12 23:17:55.846] [nas] [debug] Sending Initial Registration
[2026-04-12 23:17:55.846] [nas] [info] UE switches to state [MM-REGISTER-INITIATED]
[2026-04-12 23:17:55.846] [rrc] [debug] Sending RRC Setup Request
[2026-04-12 23:17:55.847] [rrc] [info] RRC connection established
[2026-04-12 23:17:55.847] [rrc] [info] UE switches to state [RRC-CONNECTED]
[2026-04-12 23:17:55.847] [nas] [info] UE switches to state [CM-CONNECTED]
[2026-04-12 23:18:10.861] [nas] [debug] NAS timer[3510] expired [1]
[2026-04-12 23:18:10.861] [nas] [info] UE switches to state [MM-DEREGISTERED/PS]
[2026-04-12 23:18:10.861] [nas] [info] Performing local release of NAS connection
[2026-04-12 23:18:10.861] [nas] [info] UE switches to state [MM-DEREGISTERED/ATTEMPTING-REGISTRATION]
[2026-04-12 23:18:10.861] [rrc] [info] UE switches to state [RRC-IDLE]
[2026-04-12 23:18:10.861] [nas] [info] UE switches to state [CM-IDLE]
[2026-04-12 23:18:20.871] [nas] [debug] NAS timer[3511] expired [1]
[2026-04-12 23:18:20.871] [nas] [debug] Initial registration required due to [T3511-EXPIRY-IN-ATT-REG]
[2026-04-12 23:18:20.874] [nas] [debug] UAC access attempt is allowed for identity[0], category[MO_sig]
[2026-04-12 23:18:20.874] [nas] [debug] Sending Initial Registration
[2026-04-12 23:18:20.874] [rrc] [debug] Sending RRC Setup Request
[2026-04-12 23:18:20.874] [nas] [info] UE switches to state [MM-REGISTER-INITIATED]
[2026-04-12 23:18:20.874] [rrc] [info] RRC connection established
[2026-04-12 23:18:20.874] [rrc] [info] UE switches to state [RRC-CONNECTED]
[2026-04-12 23:18:20.874] [nas] [info] UE switches to state [CM-CONNECTED]
[2026-04-12 23:18:35.886] [nas] [debug] NAS timer[3510] expired [1]
[2026-04-12 23:18:35.886] [nas] [info] UE switches to state [MM-DEREGISTERED/PS]
[2026-04-12 23:18:35.886] [nas] [info] Performing local release of NAS connection
[2026-04-12 23:18:35.886] [nas] [info] UE switches to state [MM-DEREGISTERED/ATTEMPTING-REGISTRATION]
[2026-04-12 23:18:35.886] [rrc] [info] UE switches to state [RRC-IDLE]
[2026-04-12 23:18:35.886] [nas] [info] UE switches to state [CM-IDLE]
```

Caption: Terminal logs showing UE and gNB signaling messages.

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5. Result

The simulation was partially successful. The gNB and UE were successfully configured and executed. The UE detected available signals and attempted network registration. Due to absence of a fully configured 5G core network, complete connection could not be established.

The observed logs confirm correct behavior of radio layer signaling in absence of complete core integration.

6. Conclusion

This experiment provided practical exposure to 5G network simulation using UERANSIM. Although full core network integration was not achieved, the interaction between user equipment and radio access network was successfully demonstrated through signaling logs.